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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|--------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                       | <b>NOTA TECNICA</b><br><b>TECHNICAL REPORT</b> |                          | Núm.<br>No.                                                                                 | <b>NT-T-ADP-01035</b>                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Departamento<br>Department                     | <b>Stress Methods</b>    | Avión<br>Aircraft                                                                           | <b>Technologies</b>                                            |
| <u>Titulo/Title</u><br><br><div style="text-align: center;"> <b>Program CORIN, version 3</b><br/> <b>User's Manual</b> </div>                                                                                                                                                                                                                                                                                                                                          |                                                |                          |                                                                                             |                                                                |
| <u>Palabras clave/Key words</u><br><br><b>INTERLAMINAR STRESS, SOFTWARE, COMPUTER PROGRAMS</b>                                                                                                                                                                                                                                                                                                                                                                         |                                                |                          | Clasificación acceso<br>Access class<br><br><b>P1</b>                                       | Registro de Revisiones<br>Revision Record<br><br><b>Pag. 2</b> |
| <u>Resumen/Summary</u><br><br><p>This Technical Report describes how to use the CORIN program, and includes examples to further clarify its use.</p> <p>The current program performs a bidimensional analysis of interlaminar stresses in a beam (straight or curved) of unitary width made of composite material.</p> <p>This program is included in Section XXI (Computer Programs and Procedures) of volume 8 (Stress Analysis) of the EADS CASA Design Manual.</p> |                                                |                          |                                                                                             |                                                                |
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| (*).- Cover sheet only                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Aprobado</b><br><b>Approved</b>             | Darío Fernández Villalba |                                                                                             |                                                                |

## REGISTRO DE REVISIONES/REVISIONS RECORD

[illegible]

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## 1.- Technical sheet of the program

Technical Report **NT-T-ADP-01035**

PROGRAM

CORIN

REFERENCE

DATE

April 1997

PURPOSE

This program analyses interlaminar stresses of angles and flat laminates of composite materials.

LANGUAGE

Fortran 77

COMPUTER

Sun

NUMBER OF LINES (APPROX.)

1800

EXECUTION

Interactive

AUTHOR

J.L. Cifuentes

VERSION

2.0

ADDITIONAL DOCUMENTATION

Technical Report: NT-J2-ECE-031:  
Calculation of radial stresses in angles of composite materials.

Technical Report: NT-T-ADCE-018/92:  
Theoretical Manual of the CORIN Program.  
Version 1.0

Technical Report: NT-T-ADCE-019/92:  
Programmer Manual of the CORIN Program.  
Version 1.1

## 2.- Introduction

This manual describes the use of the version 3 of the CORIN program, including examples to further clarify its use. The program performs a bidimensional analysis of interlaminar stresses in a beam made of composite material. It analyses flat laminates and angle laminates; however the program has been developed exclusively for the analysis of problems without singularities.

The beam that is being analysed is subject to transverse loads applied on one side and clamped on the other one. The program analyses interlaminar stresses that occur in an arch (1/4 circumference), in case of a curved beam and that occur in a flat laminate in case of a straight beam.

There are two versions of the program: one in the Multivac Computer and one in the PC. The use of both is explained in this manual.

The differences between this version and the previous one are:

- New PC version.
- The input data file has changed and all the data now are entered via a data file.

### 2.1 Sign Criteria

The sign criteria for the loads applied are indicated in figures 2.1 and 2.2, where the directions represented are positives.

In a straight beam, a transverse load and a bending moment can be applied. For the curved beam also a tension load and internal and external pressures are accepted. The tension load is assumed to be applied on the centre of gravity weighted by the elastic modulus on each ply, which in the case of symmetric laminates matches the mean line.

The program assumes the hypothesis of a clamped beam on one of its sides and the loads applied on the other end (isostatic problem), so that it is possible to simulate the required load conditions, at a given distance or angle, from the load application point.

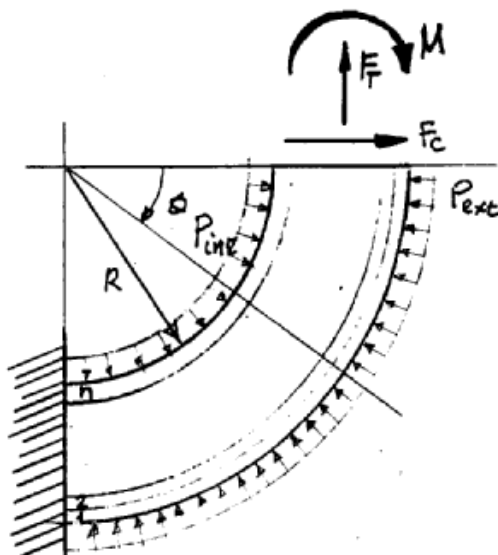


Figure 2.1. Angle beam

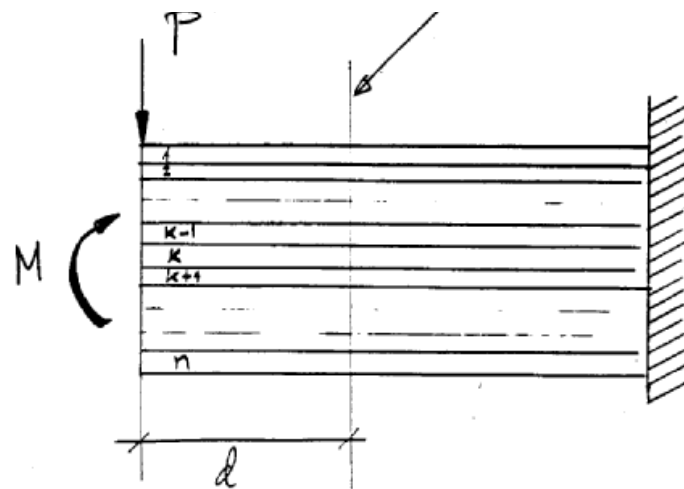


Figure 2.2. Straight beam

## 3.- Program Execution

### 3.1 PC Server Version

The Visual Basic executable program (**corin.exe**) is located in:

[\\Cae\InfoCalculo\Soft\Corin](#)

In the folder [\\Cae\InfoCalculo\Soft\Corin\Consola](#), there is another executable program, which is the Fortran program.

To execute the program, the user has to click twice on the Visual Basic executable program.

To ease this process, it is suggested creating a direct access to the program. To do so, drag with the mouse (do not copy) the **corin.exe** program file to a directory (the Desktop, for example). In this case, to execute the program, the user just has to click twice on this shortcut.

### 3.2 Multivac Computer Version

In order to execute the Multivac Program, it is necessary to previously assign the calculation symbols. The following steps must be followed:

1. Enter the following lines on the **.profile** file:

```
ENV=$HOME/.kshrc
export ENV
BASH_ENV=$HOME/.bashrc
export BASH_ENV
. asignaciones_calculo
```

2. If the user works with the **ksh** shell, the following line must be entered in the **.kshrc** file:

```
. asignaciones_calculo
```

3. If the user works with the **bash** shell, the following line must be entered in the **.bashrc** file:

```
. asignaciones_calculo
```

4. Enter the following command:

```
. .profile
```

(Or exit and enter the MULTIVAC System again)

Once this is made, the program is executed by means of the following command:

```
$corin
```

## 4.- Data Input

### 4.1 PC Server Version

In the PC version, the input data can be entered via a file or using the graphical interface. If an input file is used, it is necessary to press the 'Open' button and to choose the name of the file.

Figure 4.1. Input data

If the graphical interface is used, first the type of beam to be analysed must be chosen: straight beam or angle. The first part is the same for both types of beam. In this part, the materials number must be entered first. Then a table appears and the boxes corresponding to each material property must be filled. Below the materials properties, the number of plies must be given and another table appears where the user must choose the number of the material of each ply.

|   | $E_1$    | $E_2$  | $G_{12}$ | $\mu$ | $t$   | $\theta$ | $\varepsilon_{1t}$ | $\varepsilon_{1c}$ | $F_{1t}$ | $F_{1c}$ | $F_{2t}$ | $F_{2c}$ | ASS   | ASI  | ASR  |
|---|----------|--------|----------|-------|-------|----------|--------------------|--------------------|----------|----------|----------|----------|-------|------|------|
| 1 | 138000.0 | 7006.0 | 3700.0   | 0.30  | 0.180 | 0.0      | 1.22E-02           | -1.00E-02          | 1876.0   | 1372.0   | 71.0     | 95.0     | 100.0 | 70.0 | 30.0 |
| 2 | 138000.0 | 7006.0 | 3700.0   | 0.30  | 0.180 | 45.0     | 1.22E-02           | -1.00E-02          | 1876.0   | 1372.0   | 71.0     | 95.0     | 100.0 | 70.0 | 30.0 |
| 3 | 138000.0 | 7006.0 | 3700.0   | 0.30  | 0.180 | -45.0    | 1.22E-02           | -1.00E-02          | 1876.0   | 1372.0   | 71.0     | 95.0     | 100.0 | 70.0 | 30.0 |
| 4 | 138000.0 | 7006.0 | 3700.0   | 0.30  | 0.180 | 90.0     | 1.22E-02           | -1.00E-02          | 1876.0   | 1372.0   | 71.0     | 95.0     | 100.0 | 70.0 | 30.0 |

**Laminates:**  
 Number of plies: 10  
 Ply: 1 2 3 4 5 6 7 8 9 10  
 Material: 2 3 1 1 1 4 1 1 1 3

Figure 4.2. Material data

When all the boxes are filled, the loads and the section data are required. Both parts depend on the type of beam that has been chosen. For a straight beam, the user must give the transverse load and the



bending moment applied and the distance from the section where these are applied to the section where the calculation is required.

For an angle, the tension load, shear load and the bending moment must be entered. It is possible to give an internal and an external pressure too. Here the user can choose if the program does the analysis each 15° or in a particular section (it must be given in degrees between 0° and 90°). Moreover, in the calculation of this type of beams it is necessary to give the inner radius of the beam in order to know its geometry.

Then the user can save the data by pressing 'Save' button (the data are written on a file) or press the 'Calculation' button and wait for the results. If the 'Save' button is pressed, the program creates a data file with a '. DAT' extension that contains the data used in the analysis properly sorted. This file can be used as a data source to repeat the same analysis, or may be used as a basis for a new data file, to perform a new analysis.

## 4.2 Multivac Computer Version

Data are entered via a data file.

When the user starts the Multivac computer version of the program, the dialogue with the program is the following:

|                                                                                                                                                                                                                                                                                                                                                                                                         |  |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------|
| <p style="text-align: center;"><u>Program CORIN</u></p> <p>Interlaminar stresses analysis in composites elements.<br/>         The program performs the bidimensional stress analysis of<br/>         a beam (straight or curved) of unitary width, clamped in<br/>         one side and loaded in the other. The curved beam case<br/>         admits distributed loads (pressure) along the beam.</p> |  |            |
| Version 3.1                                                                                                                                                                                                                                                                                                                                                                                             |  | March 2013 |

To continue, press <RETURN>

### Notes

- The input data must be separated by commas and/or blanks.
- This software and the data that are produced by it are EADS-CASA property.  
 The total or partial reproduction of this software is forbidden, as well as its use out of the EADS-CASA environment.

To continue, press <RETURN>

The input data file must be introduced interactively via the keyboard:

|                             |   |
|-----------------------------|---|
| Name of the input data file | → |
|-----------------------------|---|

The full name of the file containing the output data must be given.

|                                                     |
|-----------------------------------------------------|
| Name of the output file without extension [CORIN] → |
|-----------------------------------------------------|

The program generates 3 output files:

'RES': A file containing the results of the analysis.

'LIS': A file containing the auxiliary information of the analysis.

If the user simply presses <RETURN>, the program uses the default filename 'CORIN'.

## 4.3 Input Data File

The input data file has the following characteristics:

- Everything on the right of a '!' character is assumed to be a comment. The user can add or omit as many comments as he desires.
- Data are entered in free format (separated by commas or blank).
- Any coherent unit system can be used. Nevertheless SI units are preferable and will be noted in following pages. If 'N,mm' are used the loads must be given in 'N/mm' and the moments in 'N·mm/mm'.

The input file can be divided in three groups of data: materials, laminate definition and loads. The data of each group are shown below:

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | Type of beam                                                                                                                                                                                                                                                                                                                                                                                                                    |
| MATERIALS | $\left\{ \begin{array}{l} N_m \\ i_1, E_1(1), E_2(1), G_{12}(1), \mu(1), t(1), \theta(1), \epsilon_{1t}(1), \epsilon_{1c}(1), F_{1t}(1), F_{1c}(1), F_{2t}(1), F_{2c}(1), ASS(1), ASI(1), ASR(1)^* \\ . \\ i_m, E_1(N_m), E_2(N_m), G_{12}(N_m), \mu(N_m), t(N_m), \theta(N_m), \epsilon_{1t}(N_m), \epsilon_{1c}(N_m), F_{1t}(N_m), F_{1c}(N_m), F_{2t}(N_m), F_{2c}(N_m), ASS(N_m), ASI(N_m), ASR(N_m)^* \end{array} \right.$ |
| LAMINATE  | $\left\{ \begin{array}{l} N_l \\ \text{Stacking sequence} \end{array} \right.$                                                                                                                                                                                                                                                                                                                                                  |
| LOADS     | $\left\{ \begin{array}{l} \left\{ \begin{array}{l} R, M, F_c, F_t \\ P_{int}, P_{ext} \\ \text{Option, Angle} \end{array} \right. \\ \text{or} \\ \left\{ \begin{array}{l} d, P, M \end{array} \right. \end{array} \right.$                                                                                                                                                                                                     |

\* The value of ASR is only necessary if the beam is an angle.

Next, each line of the file is explained:

**Type of beam.**- It can be: 'A' for angle or 'B' for straight beam. If 'A' is chosen, the analysis of an angle with constant inner radius will be performed. If B is chosen, the analysis of stresses between plies of a straight beam will be performed. It is irrelevant to use capital letters or not.

## **MATERIALS**

**N<sub>m</sub>.**- It is the number of materials used in the laminate. A maximum of 100 different materials can be used. Different orientations correspond to different materials. More types of plies than those to be used can be defined, but it is necessary to define every ply that is used in the description of the stacking.

**Materials average properties and allowables.**- These data must be entered for each of the different materials defined above, in free format (separated by commas or blanks). The necessary properties are:

- *i*: identifier of the material (integer number)
- $E_1, E_2, G_{12}$ : elastic moduli on the ply plane, in MPa (or N/mm<sup>2</sup>), defined on the fibre axis (in general, the orthotropic axis of the ply). Mean values will be used.
- $\mu$ : Poisson's ratio defined also on the fibre axis.
- *t*: ply thickness, in mm.
- $\theta$ : the angle formed between the axis system associated with each ply and the coordinate system of the laminate, in °.
- $\epsilon_{1t}, \epsilon_{1c}$ : allowable tension and compression strains in the fibre direction. The use of ultimate values is recommended, not taking into account the damage tolerance.
- $F_{1t}, F_{1c}$ : allowable tension and compression stresses in the fibre direction, in MPa.
- $F_{2t}, F_{2c}$ : allowable stresses in the fibre transverse direction (on its plane), in MPa.
- *ASS*: (or  $F_{12}$ ) allowable shear stress of the ply on its plane, in MPa.
- *ASI*: (or  $F_{13}$ ) allowable interlaminar shear stress, in MPa.
- *ASR*: (or  $F_{3t}$ ) allowable interlaminar tension stress, in MPa.

## **LAMINATE**

**N<sub>l</sub>.**- It's the number of plies of the laminate. A maximum of 166 plies is admitted.

**Stacking sequence.** - The stacking sequence of the laminate is provided as a sequence of numbers corresponding to the index of the identification codes of the material for each ply. The first ply in the sequence is the external ply in the case of an angle or the upper ply, according to Figure 2.2, in the case of a plane laminate, generating compression on ply 1 by positive moments.

## LOADS

The way to introduce the loads and the section where the calculation is desired is different depending on the shape of the beam.

If a curved beam is being analysed, these data must be given:

**R** : Inner radius of the curved beam, in mm.

**M** : Applied bending moment, in Nmm/mm.

**F<sub>c</sub>** : Applied shear load, in N/mm.

**F<sub>t</sub>** : Applied tension load, in N/mm.

**P<sub>int</sub>** : Pressure on the internal face of the angle, in MPa.

**P<sub>ext</sub>** : Pressure on the external face of the angle, in MPa.

**Option** : In case of a curved beam analysis, it is possible to analyse a concrete section of the arch or to make a scan of it from 0° to 90°, every 15°. For *Option* two values are possible: 1 (analysis of a determined section) or 2 (scan of sections every 15°).

**Angle** : If Option=1, it is the angle of the section to analyse in degrees. If Option=2, nothing must be written here.

If a straight beam is being analysed, these data must be given:

**d** : Distance between the section where the loads are applied and the section to analyse, in mm.

**P** : Applied shear load, in N/mm.

**M** : Applied bending moment, in Nmm/mm.

NOTE: The signs criteria for the loads are shown on figures 2.1 and 2.2.

## 5.- Results and Output files

### 5.1 PC Server Version

In the PC version, when all data is entered and the button 'Calculate' is pressed, the results calculated by the program are shown:

**Results of plies with minimum RF:**

Section: 0

|                            | Ply | Fibre Strain | Tang. Stress | Radial Stress | Transv. Shear | RF   |
|----------------------------|-----|--------------|--------------|---------------|---------------|------|
| <b>Maximum Strain</b>      | 12  | 4.172E-03    | 575.79       | -0.30         | 0.00          | >2.0 |
| <b>Max. Stress (Hoff.)</b> | 1   | 1.014E-03    | -57.00       | 1.49          | -2.08         | >2.0 |
| <b>Delamination</b>        | 7   | 1.083E-04    | 14.95        | 24.04         | -37.16        | 1.04 |

**Results ply by ply:**

Section: 0 Ply: 1

| Stresses and Strains:   |           | Reserve Factors:          |       |
|-------------------------|-----------|---------------------------|-------|
| Fibre Strain .....      | 1.014E-03 | Maximum Strain .....      | 12.03 |
| Tang. Stress .....      | -57.00    | Max. Stress (Hoff.) ..... | 4.96  |
| Radial Stress .....     | 1.49      | Delamination .....        | 17.31 |
| Transv. Shear Stress .. | -2.08     |                           |       |

Write .RES  
Write .LIS  
Return to data  
End of the program

Figure 5.1. Results form

There are two different parts in the form. The first one shows the ply with the minimum reserve factor for each failure criteria, and the strain and stresses corresponding to that ply. If an analysis each 15° has been chosen, the user can see the results for each section by choosing the value (in degrees) of the section. The reserve factors less than 1 are written in red.

The second part shows the results for each ply of the section given (or for each 15° like in the table before). The user can choose the ply to see. In the two cases, if the beam is an angle, the radial stresses are also calculated.

On the right, there are four buttons. The first and the second permit the user to write the results on two different files: one with the results in the plies with minimum reserve factors ('.RES') and one with the results of all plies ('.LIS'). The configurations of both files are explained in the next paragraph. The other two buttons are to return to the data and to finish the program.

## **5.2 Multivac Computer Version**

Every time the program is run, it generates two files, named:

`<Name>.RESand    <Name>.LIS`

where `<Name>` is the filename that was entered when the program requested a name for the output files. The default filename is 'CORIN'.

## 5.3 Output Files

The content of the output files is the following:

1)

|                 |
|-----------------|
| File <Name>.RES |
|-----------------|

In this file the input data and a summary of the critical reserve factors according to the different failure criteria are included.

### INPUT DATA

The material properties and the allowables of each ply are given:

**Material properties.** - For each ply the following properties are given:

- PLY: Ply index. The first ply is the external ply.
- E1, E2, G12, NU12: Elastic moduli in longitudinal and transverse direction, shear modulus and Poisson ratio of the material of the ply.
- THICKNESS: Thickness of the ply.
- ANGLE: Angle formed between the axis system associated with each ply and the coordinate system of the laminate.
- MEAN RADIUS: Radial co-ordinate of the mean line of the ply, measured from the centre of the angle inner radius (only for curved beams).

**Ply allowables.** - For each ply the following allowables are given:

- PLY: Ply index. The first ply is the external ply.
- T.STRAIN, C.STRAIN: allowable tension and compression strains in the fibre direction.
- F1T, F1C, F2T, F2C: allowable tension and compression stresses in the fibre direction and in the fibre transverse direction (on its plane).
- ASS: Stress in the tangent of the angle. It corresponds to tension-compression stresses produced by bending and axial force. They are provided on the mean line of the ply. (+ Tension).
- ASI: Shear stress between laminates. Provided values correspond to the internal /lower interply of the laminate.
- ASR: It corresponds to the radial stress (perpendicular tension of the laminate). The results provided are given on the internal interply of each laminate and only for curved beams.



**Total thickness and loads applied.** - The total thickness calculated by the program and the loads applied given in the input data file are written after the ply allowables. The internal radius is also given if the beam is an angle.

#### SECTION MINIMUM RESERVE FACTORS

For a straight beam the minimum reserve factors in the section defined in the input data are given. For a curved beam, if a determined section was given, these reserve factors are calculated only for this angle. But if a scan of sections was chosen, the program obtains the minimum reserve factor every 15° section.

The reserve factors provided correspond to the following failure criteria:

- MAX. STRAIN: Maximum fibre strain criteria.
- HOFFMAN: Modified Hoffman stresses criteria. It corresponds to the following equation:

$$\left( \frac{\sigma_{11}^2}{F_{1t}F_{1c}} \right) - \left( \frac{\sigma_{11}\sigma_{22}}{F_{1t}F_{1c}} \right) + \left( \frac{\sigma_{22}^2}{F_{2t}F_{2c}} \right) + \left( \frac{\sigma_{12}}{A_{SS}} \right)^2 = e^2$$

$$R.F. = 1/e$$

- DELAMINATION: Quadratic failure criteria. The combination is only made for positive radial stresses (tension). For the compression radial stresses, the interlaminar shear is compared with its allowable. The interply failure due to radial compression stresses occurs not very often. Moreover nor the allowable stress, nor the failure mechanism, neither the possible combination with the shear failure are available.

$$\left( \frac{\sigma_r}{A_{SR}} \right)^2 + \left( \frac{\sigma_{rt}}{A_{SI}} \right)^2 = d^2$$

$$R.F. = 1/d$$

2)

File <Name>.LIS

In this file for each section analysed and for each ply the following results are given:

- PLY MEAN RADIUS (only for curved beams).
- TANGENCIAL, RADIAL AND SHEAR STRESSES at each ply of the section.
- RESERVE FACTORS: They are calculated in the same way as it has been shown before.

To facilitate the location in the .LIS file of the ply where any type of failure occurs, messages beginning and ending with three asterisks are displayed.

## **5.4 Warning Messages**

Related to the possible errors in the input data file, the program can display the following advices and then the program is stopped:

\*\*\* INCORRECT TYPE OF BEAM \*\*\*

\*\*\* INCORRECT MATERIALS NUMBER \*\*\*

\*\*\* THE MAXIMUM NUMBER OF MATERIALS IS 100 \*\*\*

\*\*\* ERROR IN THE MATERIALS PROPERTIES \*\*\*

\*\*\* INCORRECT PLIES NUMBER \*\*\*

\*\*\* THE NUMBER MAXIMUM OF PLIES IS 166 \*\*\*

\*\*\* ERROR IN THE STACKING SEQUENCE \*\*\*

\*\*\* ERROR IN THE APPLIED LOADS \*\*\*

\*\*\* THE INNER RADIUS CANNOT BE NEGATIVE \*\*\*

\*\*\* INCORRECT SECTION DEFINITON OPTION \*\*\*

\*\*\* THE ANGLE SHOULD BE BETWEEN 0° AND 90° \*\*\*

During the execution of the program, if the matrix associated to the equation system is not regular the following error messages can be displayed:

\*\*\* ERROR: SINGULAR MATRIX \*\*\*

It might be an error in the input properties or an incongruent value in the dimensions. The program is stopped.

## **6.- Applicability and Limitations**

- a) The CORIN program is based on a bidimensional stress analysis, so it performs the analysis of a beam (straight or curved) with unitary width, under the hypothesis that the conditions do not change along that dimension (see Fig. 2.1 and 2.2). The problems where an important three-dimensional effect might occur cannot be analysed directly.
- b) The program assumes that the beam is clamped on one end, free on the other and loaded on the free end. In general, the actual problem to solve will have different support conditions, so the user must calculate previously the load conditions equivalent to the conditions considered by the program. This is very important because otherwise the results might be totally different from those expected.
- c) Problems with singularities (free edges, holes, previous delaminations, etc.) or problems where the time is considered (growth of delaminations or damages, fatigue, etc.) are not analysed.
- d) The maximum number of materials defined is 100.
- e) The maximum number of plies on a stack to analyse is 166.

## 7.- Examples

Below there are 3 examples of the use of this program. It is hoped that these examples will clarify the information given in this manual.

**EXAMPLE 1**

In this example a curved beam is analysed. The results are wanted in a determined section which angle is  $10^\circ$ . The laminate is made of a material whose properties are:

$$\begin{aligned} E_1 &= 138000 \text{ N/mm}^2 & \mu &= 0.3 \\ E_2 &= 7006 \text{ N/mm}^2 & t &= 0.18 \text{ mm} \\ G_{12} &= 3700 \text{ N/mm}^2 \end{aligned}$$

And whose allowables are:

$$\begin{aligned} \epsilon_{1t} &= 12174 \mu \epsilon & F_{1t} &= 1876 \text{ N/mm}^2 & ASS &= 100 \text{ N/mm}^2 \\ \epsilon_{1c} &= -10000 \mu \epsilon & F_{2t} &= 71 \text{ N/mm}^2 & ASI &= 70 \text{ N/mm}^2 \\ & & F_{1c} &= 1372 \text{ N/mm}^2 & ASR &= 30 \text{ N/mm}^2 \\ & & F_{2c} &= 95 \text{ N/mm}^2 \end{aligned}$$

The lay-up is: (45/-45/0/0/0/90/0/0/0/-45/45/0/0/45/-45/0/0/0/90/0/0/0/-45/45) and the beam inner radius is 5 mm. The applied loads are a shear load of 100 N/mm, a tension load of 100 N/mm, a bending moment of 400 N·mm/mm, an internal pressure of 0.3 N/mm<sup>2</sup> and an external pressure of 0.01 N/mm<sup>2</sup>.

The input file is:

File ej\_corin\_1.dat

-----

A

4

1,138000,7006,3700,.3,.18,0.,12174e-6,-10000e-6,1876,1372,71,95,100,70,30

2,138000,7006,3700,.3,.18,45.,12174e-6,-10000e-6,1876,1372,71,95,100,70,30

3,138000,7006,3700,.3,.18,-45.,12174e-6,-10000e-6,1876,1372,71,95,100,70,30

4,138000,7006,3700,.3,.18,90.,12174e-6,-10000e-6,1876,1372,71,95,100,70,30

24

2,3,1,1,1,4,1,1,1,3,2,1,1,2,3,1,1,1,4,1,1,1,3,2

5,400,100,100

.3,.01

1,10

The output files are:

File ej\_corin\_1.res

-----

# INPUT DATA

=====

## MATERIAL PROPERTIES:

| PLY | E1      | E2    | G12   | NU12 | THICKNESS | ANGLE | MEAN RADIUS |
|-----|---------|-------|-------|------|-----------|-------|-------------|
| 1   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 45.0  | 5.090       |
| 2   | 138000. | 7006. | 3700. | 0.30 | 0.180     | -45.0 | 5.270       |
| 3   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 5.450       |
| 4   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 5.630       |
| 5   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 5.810       |
| 6   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 90.0  | 5.990       |
| 7   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 6.170       |
| 8   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 6.350       |
| 9   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 6.530       |
| 10  | 138000. | 7006. | 3700. | 0.30 | 0.180     | -45.0 | 6.710       |
| 11  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 45.0  | 6.890       |
| 12  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 7.070       |
| 13  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 7.250       |
| 14  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 45.0  | 7.430       |
| 15  | 138000. | 7006. | 3700. | 0.30 | 0.180     | -45.0 | 7.610       |
| 16  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 7.790       |
| 17  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 7.970       |
| 18  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 8.150       |
| 19  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 90.0  | 8.330       |
| 20  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 8.510       |
| 21  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 8.690       |
| 22  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 8.870       |
| 23  | 138000. | 7006. | 3700. | 0.30 | 0.180     | -45.0 | 9.050       |
| 24  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 45.0  | 9.230       |

## PLY ALLOWABLES:

| PLY | T.STRAIN  | C.STRAIN  | F1T   | F1C   | F2T | F2C | ASS  | ASI | ASR |
|-----|-----------|-----------|-------|-------|-----|-----|------|-----|-----|
| 1   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 2   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 3   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 4   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 5   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 6   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 7   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 8   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 9   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 10  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 11  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 12  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 13  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 14  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 15  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 16  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 17  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 18  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 19  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 20  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 21  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 22  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 23  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 24  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |

INTERNAL RADIUS = 5.000  
TOTAL THICKNESS = 4.320

APPLIED LOADS:

BENDING MOMENT = 400.0  
SHEAR LOAD = 100.0  
TENSION LOAD = 100.0  
INTERNAL PRESS. = 0.300  
EXTERNAL PRESS. = 0.010

SECTION MINIMUM RESERVE FACTORS:

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SECTION AT 10.0°

| CRITERIA      | PLY | FIBRE<br>STRAIN | TANGENTIAL<br>STRESS | RADIAL<br>STRESS | TRANSV. SHEAR<br>STRESS | R.F. |
|---------------|-----|-----------------|----------------------|------------------|-------------------------|------|
| MAX. STRAIN   | 22  | 0.2533E-02      | 349.5                | 3.35             | -3.55                   | >2.0 |
| STRESS(HOFF.) | 24  | -.1101E-02      | 61.9                 | -0.30            | 0.00                    | >2.0 |
| DELAMINATION  | 15  | -.3056E-03      | 17.2                 | 27.99            | -32.40                  | 0.96 |

If the PC version is used and the input file 'ej\_corin\_1.dat', the program shows the following on the screen:

**Beam type:** ☐ Straight Beam ☒ Angle

Inner radius ..... 5.00

**Materials:**

Number of materials: 4

|   | E <sub>1</sub> | E <sub>2</sub> | G <sub>12</sub> | μ    | t     | θ     | ε <sub>1t</sub> | ε <sub>1c</sub> | F <sub>1t</sub> | F <sub>1c</sub> | F <sub>2t</sub> | F <sub>2c</sub> | ASS   | ASI  | ASR  |
|---|----------------|----------------|-----------------|------|-------|-------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-------|------|------|
| 1 | 138000.0       | 7006.0         | 3700.0          | 0.30 | 0.180 | 0.0   | 1.22E-02        | -1.00E-02       | 1876.0          | 1372.0          | 71.0            | 95.0            | 100.0 | 70.0 | 30.0 |
| 2 | 138000.0       | 7006.0         | 3700.0          | 0.30 | 0.180 | 45.0  | 1.22E-02        | -1.00E-02       | 1876.0          | 1372.0          | 71.0            | 95.0            | 100.0 | 70.0 | 30.0 |
| 3 | 138000.0       | 7006.0         | 3700.0          | 0.30 | 0.180 | -45.0 | 1.22E-02        | -1.00E-02       | 1876.0          | 1372.0          | 71.0            | 95.0            | 100.0 | 70.0 | 30.0 |
| 4 | 138000.0       | 7006.0         | 3700.0          | 0.30 | 0.180 | 90.0  | 1.22E-02        | -1.00E-02       | 1876.0          | 1372.0          | 71.0            | 95.0            | 100.0 | 70.0 | 30.0 |

**Laminates:**

Number of plies: 24

| Ply | Material |
|-----|----------|
| 1   | 1        |
| 2   | 3        |
| 3   | 2        |
| 4   | 1        |
| 5   | 1        |
| 6   | 2        |
| 7   | 3        |
| 8   | 1        |
| 9   | 1        |
| 10  | 1        |
| 11  | 4        |
| 12  | 1        |
| 13  | 1        |
| 14  | 1        |
| 15  | 3        |
| 16  | 2        |

**Applied loads:**

Bending Moment ..... 400.00

Shear Load ..... 100.00

Tension Load ..... 100.00

Inside Pressure ..... 0.30000

Outside Pressure ..... 0.01000

**Section:**

☒ Given angle (\*) 10.0

☐ Each 15°

Calculate

Figure 7.1. Input data of the example 1

If the 'Calculate' button is pressed the results are shown:

**RESULTS OF PLYS WITH MINIMUM RF**

Section: 10°

|                     | Ply | Fibre Strain | Tang. Stress | Radial Stress | Transv. Shear | RF   |
|---------------------|-----|--------------|--------------|---------------|---------------|------|
| Maximum Strain      | 22  | 2.533E-03    | 349.53       | 3.35          | -3.55         | >2.0 |
| Max. Stress (Hoff.) | 24  | -1.101E-03   | 61.89        | -0.30         | 0.00          | >2.0 |
| Delamination        | 15  | -3.056E-04   | 17.18        | 28.00         | -32.40        | 0.96 |

**RESULTS PLY BY PLY**

Section: 10° Ply: 15

| Stresses and Strains: |            | Reserve Factors:    |       |
|-----------------------|------------|---------------------|-------|
| Fibre Strain          | -3.056E-04 | Maximum Strain      | 32.72 |
| Tang. Stress          | 17.18      | Max. Stress (Hoff.) | 7.39  |
| Radial Stress         | 28.00      | Delamination        | 0.96  |
| Transv. Shear Stress  | -32.40     |                     |       |

Write .RES

Write .LIS

Return to data

End of the program

Figure 7.2. Output data of the example 1



|           |
|-----------|
| EXAMPLE 2 |
|-----------|

In this example the beam to analysis is the same as in the example 1 but the loads applied are different: a shear load of 100 N/mm, a tension load of 100 N/mm and a bending moment of 500 N·mm/mm. Neither internal nor external pressures are applied. A scan of the arch from 0° to 90° is required.

The input file is:

File ej\_corin\_2.dat

-----

A

4

1,138000,7006,3700,.3,.18,0.,12174e-6,-10000e-6,1876,1372,71,95,100,70,30

2,138000,7006,3700,.3,.18,45.,12174e-6,-10000e-6,1876,1372,71,95,100,70,30

3,138000,7006,3700,.3,.18,-45.,12174e-6,-10000e-6,1876,1372,71,95,100,70,30

4,138000,7006,3700,.3,.18,90.,12174e-6,-10000e-6,1876,1372,71,95,100,70,30

24

2,3,1,1,1,4,1,1,1,3,2,1,1,2,3,1,1,1,4,1,1,1,3,2

5,500,100,100

0,0

2

The output files are:

File ej\_corin\_2.res

-----

# INPUT DATA

=====

## MATERIAL PROPERTIES:

| PLY | E1      | E2    | G12   | NU12 | THICKNESS | ANGLE | MEAN RADIUS |
|-----|---------|-------|-------|------|-----------|-------|-------------|
| 1   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 45.0  | 5.090       |
| 2   | 138000. | 7006. | 3700. | 0.30 | 0.180     | -45.0 | 5.270       |
| 3   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 5.450       |
| 4   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 5.630       |
| 5   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 5.810       |
| 6   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 90.0  | 5.990       |
| 7   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 6.170       |
| 8   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 6.350       |
| 9   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 6.530       |
| 10  | 138000. | 7006. | 3700. | 0.30 | 0.180     | -45.0 | 6.710       |
| 11  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 45.0  | 6.890       |
| 12  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 7.070       |
| 13  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 7.250       |
| 14  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 45.0  | 7.430       |
| 15  | 138000. | 7006. | 3700. | 0.30 | 0.180     | -45.0 | 7.610       |
| 16  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 7.790       |
| 17  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 7.970       |
| 18  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 8.150       |
| 19  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 90.0  | 8.330       |
| 20  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 8.510       |
| 21  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 8.690       |
| 22  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   | 8.870       |
| 23  | 138000. | 7006. | 3700. | 0.30 | 0.180     | -45.0 | 9.050       |
| 24  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 45.0  | 9.230       |

## PLY ALLOWABLES:

| PLY | T.STRAIN  | C.STRAIN  | F1T   | F1C   | F2T | F2C | ASS  | ASI | ASR |
|-----|-----------|-----------|-------|-------|-----|-----|------|-----|-----|
| 1   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 2   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 3   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 4   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 5   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 6   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 7   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 8   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 9   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 10  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 11  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 12  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 13  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 14  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 15  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 16  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 17  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 18  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 19  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 20  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 21  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 22  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 23  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |
| 24  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. | 30. |

INTERNAL RADIUS = 5.000  
TOTAL THICKNESS = 4.320

APPLIED LOADS:  
BENDING MOMENT = 500.0  
SHEAR LOAD = 100.0  
TENSION LOAD = 100.0  
INTERNAL PRESS. = 0.000  
EXTERNAL PRESS. = 0.000

# SECTION MINIMUM RESERVE FACTORS:

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## SECTION AT 0.0°

| CRITERIA      | PLY | FIBRE<br>STRAIN | TANGENTIAL<br>STRESS | RADIAL<br>STRESS | TRANSV. SHEAR<br>STRESS | R.F. |
|---------------|-----|-----------------|----------------------|------------------|-------------------------|------|
| MAX. STRAIN   | 22  | 0.2410E-02      | 332.5                | 3.40             | -4.37                   | >2.0 |
| STRESS(HOFF.) | 24  | -.1027E-02      | 57.7                 | 0.00             | 0.00                    | >2.0 |
| DELAMINATION  | 15  | -.2868E-03      | 16.1                 | 27.50            | -39.83                  | 0.93 |

\*\*\*\*\*

## SECTION AT 15.0°

| CRITERIA      | PLY | FIBRE<br>STRAIN | TANGENTIAL<br>STRESS | RADIAL<br>STRESS | TRANSV. SHEAR<br>STRESS | R.F. |
|---------------|-----|-----------------|----------------------|------------------|-------------------------|------|
| MAX. STRAIN   | 22  | 0.3141E-02      | 433.4                | 4.38             | -3.09                   | >2.0 |
| STRESS(HOFF.) | 24  | -.1320E-02      | 74.2                 | 0.00             | 0.00                    | 1.71 |
| DELAMINATION  | 15  | -.3703E-03      | 20.8                 | 36.45            | -28.16                  | 0.78 |

\*\*\*\*\*

## SECTION AT 30.0°

| CRITERIA      | PLY | FIBRE<br>STRAIN | TANGENTIAL<br>STRESS | RADIAL<br>STRESS | TRANSV. SHEAR<br>STRESS | R.F. |
|---------------|-----|-----------------|----------------------|------------------|-------------------------|------|
| MAX. STRAIN   | 22  | 0.3600E-02      | 496.9                | 5.00             | -1.60                   | >2.0 |
| STRESS(HOFF.) | 24  | -.1503E-02      | 84.5                 | 0.00             | 0.00                    | 1.50 |
| DELAMINATION  | 15  | -.4227E-03      | 23.8                 | 42.08            | -14.58                  | 0.71 |

\*\*\*\*\*

## SECTION AT 45.0°

| CRITERIA      | PLY | FIBRE<br>STRAIN | TANGENTIAL<br>STRESS | RADIAL<br>STRESS | TRANSV. SHEAR<br>STRESS | R.F. |
|---------------|-----|-----------------|----------------------|------------------|-------------------------|------|
| MAX. STRAIN   | 22  | 0.3757E-02      | 518.5                | 5.21             | 0.00                    | >2.0 |
| STRESS(HOFF.) | 24  | -.1566E-02      | 88.0                 | 0.00             | 0.00                    | 1.44 |
| DELAMINATION  | 15  | -.4406E-03      | 24.8                 | 44.00            | 0.00                    | 0.68 |

\*\*\*\*\*

## SECTION AT 60.0°

| CRITERIA    | PLY | FIBRE<br>STRAIN | TANGENTIAL<br>STRESS | RADIAL<br>STRESS | TRANSV. SHEAR<br>STRESS | R.F. |
|-------------|-----|-----------------|----------------------|------------------|-------------------------|------|
| MAX. STRAIN | 22  | 0.3600E-02      | 496.9                | 5.00             | 1.60                    | >2.0 |

|               |    |            |      |       |       |      |
|---------------|----|------------|------|-------|-------|------|
| STRESS(HOFF.) | 24 | -.1503E-02 | 84.5 | 0.00  | 0.00  | 1.50 |
| DELAMINATION  | 15 | -.4227E-03 | 23.8 | 42.08 | 14.58 | 0.71 |

\*\*\*\*\*

SECTION AT 75.0°

| CRITERIA      | PLY | FIBRE<br>STRAIN | TANGENTIAL<br>STRESS | RADIAL<br>STRESS | TRANSV.SHEAR<br>STRESS | R.F. |
|---------------|-----|-----------------|----------------------|------------------|------------------------|------|
| MAX.STRAIN    | 22  | 0.3141E-02      | 433.4                | 4.38             | 3.09                   | >2.0 |
| STRESS(HOFF.) | 24  | -.1320E-02      | 74.2                 | 0.00             | 0.00                   | 1.71 |
| DELAMINATION  | 15  | -.3703E-03      | 20.8                 | 36.45            | 28.16                  | 0.78 |

\*\*\*\*\*

SECTION AT 90.0°

| CRITERIA      | PLY | FIBRE<br>STRAIN | TANGENTIAL<br>STRESS | RADIAL<br>STRESS | TRANSV.SHEAR<br>STRESS | R.F. |
|---------------|-----|-----------------|----------------------|------------------|------------------------|------|
| MAX.STRAIN    | 22  | 0.2410E-02      | 332.5                | 3.40             | 4.37                   | >2.0 |
| STRESS(HOFF.) | 24  | -.1027E-02      | 57.7                 | 0.00             | 0.00                   | >2.0 |
| DELAMINATION  | 15  | -.2868E-03      | 16.1                 | 27.50            | 39.83                  | 0.93 |

If the PC version is used and the input file 'ej\_corin\_2.dat', the program shows the following on the screen:

**Corin**

**Beam type:** ☐ Straight Beam ☒ Angle

Inner radius ..... 5.00

**Materials:**

Number of materials: 4

|   | E <sub>1</sub> | E <sub>2</sub> | G <sub>12</sub> | μ    | t     | θ     | ε <sub>1t</sub> | ε <sub>1c</sub> | F <sub>1t</sub> | F <sub>1c</sub> | F <sub>2t</sub> | F <sub>2c</sub> | ASS   | ASI  | ASR  |
|---|----------------|----------------|-----------------|------|-------|-------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-------|------|------|
| 1 | 138000.0       | 7006.0         | 3700.0          | 0.30 | 0.180 | 0.0   | 1.22E-02        | -1.00E-02       | 1876.0          | 1372.0          | 71.0            | 95.0            | 100.0 | 70.0 | 30.0 |
| 2 | 138000.0       | 7006.0         | 3700.0          | 0.30 | 0.180 | 45.0  | 1.22E-02        | -1.00E-02       | 1876.0          | 1372.0          | 71.0            | 95.0            | 100.0 | 70.0 | 30.0 |
| 3 | 138000.0       | 7006.0         | 3700.0          | 0.30 | 0.180 | -45.0 | 1.22E-02        | -1.00E-02       | 1876.0          | 1372.0          | 71.0            | 95.0            | 100.0 | 70.0 | 30.0 |
| 4 | 138000.0       | 7006.0         | 3700.0          | 0.30 | 0.180 | 90.0  | 1.22E-02        | -1.00E-02       | 1876.0          | 1372.0          | 71.0            | 95.0            | 100.0 | 70.0 | 30.0 |

**Laminates:**

Number of plies: 24

Ply: 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24

Material: 1 3 2 1 1 2 3 1 1 1 4 1 1 1 3 2

**Applied loads:**

Bending Moment ..... 500.00

Shear Load ..... 100.00

Tension Load ..... 100.00

Inside Pressure ..... 0.00000

Outside Pressure ..... 0.00000

**Section:**

☐ Given angle (°)

☒ Each 15°

Calculate

Figure 7.3. Input data of the example 2

If the 'Calculate' button is pressed the results are shown:

**Corin**

**RESULTS OF PLIES WITH MINIMUM RF**

Section: 0

|                     | Ply | Fibre Strain | Tang. Stress | Radial Stress | Transv. Shear | RF   |
|---------------------|-----|--------------|--------------|---------------|---------------|------|
| Maximum Strain      | 22  | 2.410E-03    | 332.52       | 3.40          | -4.37         | >2.0 |
| Max. Stress (Hoff.) | 24  | -1.027E-03   | 57.75        | 0.00          | 0.00          | >2.0 |
| Delamination        | 15  | -2.868E-04   | 16.12        | 27.50         | -39.83        | 0.93 |

**RESULTS PLY BY PLY**

Section: 0 Ply: 15

**Stresses and Strains:**

Fibre Strain ..... -2.868E-04

Tang. Stress ..... 16.12

Radial Stress ..... 27.50

Transv. Shear Stress ..... -39.83

**Reserve Factors:**

Maximum Strain ..... 34.87

Max. Stress (Hoff.) ..... 7.87

Delamination ..... 0.93

Write .RES

Write .LIS

Return to data

End of the program

Figure 7.4. Output data of the example 2

|           |
|-----------|
| EXAMPLE 3 |
|-----------|

In this example a straight beam is analysed. It is made of the same material and lay-up as in the previous examples. The results are required in a section placed 5 mm from the clamped edge of the beam and the applied loads are a bending moment of 400 Nmm/mm and a shear load of 100N/mm.

The input file is:

File ej\_corin\_3.dat

-----

B

4

1,138000,7006,3700,.3,.18,0.,12174e-6,-10000e-6,1876,1372,71,95,100,70,30

2,138000,7006,3700,.3,.18,45.,12174e-6,-10000e-6,1876,1372,71,95,100,70,30

3,138000,7006,3700,.3,.18,-45.,12174e-6,-10000e-6,1876,1372,71,95,100,70,30

4,138000,7006,3700,.3,.18,90.,12174e-6,-10000e-6,1876,1372,71,95,100,70,30

24

2,3,1,1,1,4,1,1,1,3,2,1,1,2,3,1,1,1,4,1,1,1,3,2

5,400,100

The output files are:

File ej\_corin\_3.res

-----

# INPUT DATA

=====

## MATERIAL PROPERTIES:

| PLY | E1      | E2    | G12   | NU12 | THICKNESS | ANGLE |
|-----|---------|-------|-------|------|-----------|-------|
| 1   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 45.0  |
| 2   | 138000. | 7006. | 3700. | 0.30 | 0.180     | -45.0 |
| 3   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   |
| 4   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   |
| 5   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   |
| 6   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 90.0  |
| 7   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   |
| 8   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   |
| 9   | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   |
| 10  | 138000. | 7006. | 3700. | 0.30 | 0.180     | -45.0 |
| 11  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 45.0  |
| 12  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   |
| 13  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   |
| 14  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 45.0  |
| 15  | 138000. | 7006. | 3700. | 0.30 | 0.180     | -45.0 |
| 16  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   |
| 17  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   |
| 18  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   |
| 19  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 90.0  |
| 20  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   |
| 21  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   |
| 22  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 0.0   |
| 23  | 138000. | 7006. | 3700. | 0.30 | 0.180     | -45.0 |
| 24  | 138000. | 7006. | 3700. | 0.30 | 0.180     | 45.0  |

## PLY ALLOWABLES:

| PLY | T.STRAIN  | C.STRAIN  | F1T   | F1C   | F2T | F2C | ASS  | ASI |
|-----|-----------|-----------|-------|-------|-----|-----|------|-----|
| 1   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 2   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 3   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 4   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 5   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 6   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 7   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 8   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 9   | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 10  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 11  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 12  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 13  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 14  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 15  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 16  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 17  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 18  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 19  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 20  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 21  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 22  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 23  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |
| 24  | 0.122E-01 | -.100E-01 | 1876. | 1372. | 71. | 95. | 100. | 70. |

TOTAL THICKNESS = 4.320

APPLIED LOADS:

BENDING MOMENT = 100.0

SHEAR LOAD = 400.0

SECTION MINIMUM RESERVE FACTORS:

=====

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SECTION AT 5.0

| CRITERIA      | PLY | FIBRE<br>STRAIN | TENSILE<br>STRESS | TRANSV. SHEAR<br>STRESS | R.F. |
|---------------|-----|-----------------|-------------------|-------------------------|------|
| MAX. STRAIN   | 22  | -.6532E-02      | -901.4            | -7.72                   | 1.53 |
| STRESS(HOFF.) | 1   | -.1894E-02      | 106.5             | -4.04                   | 1.19 |
| DELAMINATION  | 12  | 0.3438E-03      | 47.4              | -152.34                 | 0.46 |



If the PC version is used and the input file 'ej\_corin\_3.dat', the program shows the following on the screen:

**Beam type:** ☒ Straight Beam ☐ Angle

**Materials:**  
Number of materials: 4

|   | $E_1$    | $E_2$  | $G_{12}$ | $\mu$ | $t$   | $\theta$ | $\varepsilon_{1t}$ | $\varepsilon_{1c}$ | $F_{1t}$ | $F_{1c}$ | $F_{2t}$ | $F_{2c}$ | ASS   | ASI  |
|---|----------|--------|----------|-------|-------|----------|--------------------|--------------------|----------|----------|----------|----------|-------|------|
| 1 | 138000.0 | 7006.0 | 3700.0   | 0.30  | 0.180 | 0.0      | 1.22E-02           | -1.00E-02          | 1876.0   | 1372.0   | 71.0     | 95.0     | 100.0 | 70.0 |
| 2 | 138000.0 | 7006.0 | 3700.0   | 0.30  | 0.180 | 45.0     | 1.22E-02           | -1.00E-02          | 1876.0   | 1372.0   | 71.0     | 95.0     | 100.0 | 70.0 |
| 3 | 138000.0 | 7006.0 | 3700.0   | 0.30  | 0.180 | -45.0    | 1.22E-02           | -1.00E-02          | 1876.0   | 1372.0   | 71.0     | 95.0     | 100.0 | 70.0 |
| 4 | 138000.0 | 7006.0 | 3700.0   | 0.30  | 0.180 | 90.0     | 1.22E-02           | -1.00E-02          | 1876.0   | 1372.0   | 71.0     | 95.0     | 100.0 | 70.0 |

**Laminates:**  
Number of plies: 24

| Ply | Material |
|-----|----------|
| 9   | 1        |
| 10  | 3        |
| 11  | 2        |
| 12  | 1        |
| 13  | 1        |
| 14  | 2        |
| 15  | 3        |
| 16  | 1        |
| 17  | 1        |
| 18  | 1        |
| 19  | 4        |
| 20  | 1        |
| 21  | 1        |
| 22  | 1        |
| 23  | 3        |
| 24  | 2        |

**Applied loads:**  
Shear load ..... 400.00  
Bending Moment ..... 100.00

**Section:**  
Distance to the load ..... 5.00

**Calculate**

File icons: [New] [Open] [Save] [Print] [Language: ES, EN]

Figure 7.5. Input data of the example 3

If the 'Calculate' button is pressed the results are shown:

**RESULTS OF PLYS WITH MINIMUM RF**  
Section: 5

|                            | Ply | Fibre Strain | Tang. Stress | Transv. Shear | RF   |
|----------------------------|-----|--------------|--------------|---------------|------|
| <b>Maximum Strain</b>      | 22  | -6.532E-03   | -901.41      | -7.72         | 1.53 |
| <b>Max. Stress (Hoff.)</b> | 1   | -1.894E-03   | 106.51       | -4.04         | 1.19 |
| <b>Delamination</b>        | 12  | 3.438E-04    | 47.44        | -152.34       | 0.46 |

**RESULTS PLY BY PLY**  
Section: 5 Ply: 12

| Stresses and Strains:   |           | Reserve Factors:          |       |
|-------------------------|-----------|---------------------------|-------|
| Fibre Strain .....      | 3.438E-04 | Maximum Strain .....      | 35.49 |
| Tang. Stress .....      | 47.44     | Max. Stress (Hoff.) ..... | 33.82 |
| Transv. Shear Stress .. | -152.34   | Delamination .....        | 0.46  |

Write .RES  
Write .LIS  
Return to data  
End of the program

Figure 7.6. Output data of the example 3